

Same-Weight Transport and the Weighted Hardy Bottleneck

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Source Prompt

The source is Tomasz Kiwerski and Jakub Tomaszewski, *Arithmetic, interpolation and factorization of amalgams*, arXiv:2401.05526. In Section 8.b, page 76, the authors ask for abstract analogues of weighted factorization results for generalized weighted spaces X_w .

The result below gives such an analogue for the common-weight transport case. It is a partial result: it does not address different weights, later symmetrization, or the separate Orlicz-Cesaro multiplier problem.

Definitions

Let $I = (0, \infty)$. Let w be a positive weight and set

$$W(t) = \int_0^t w(s) ds.$$

Assume $W(t) < \infty$ for every $t < \infty$ and $W(\infty) = \infty$, so that W has an increasing inverse $\tau = W^{-1}$ on I .

For a Banach ideal space X on I , define its transported weighted copy

$$X^{[w]} = \{f : f \circ \tau \in X\}, \quad \|f\|_{X^{[w]}} = \|f \circ \tau\|_X.$$

When X is rearrangement invariant, this agrees isometrically with the source paper's generalized weighted space X_w . The notation $X^{[w]}$ is used only to emphasize that the proof needs the multiplicative transport map $T_w f = f \circ \tau$, and hence also applies to non-rearrangement-invariant Banach ideal spaces such as Cesaro or Tandori-type spaces.

For Banach ideal spaces X, Y , write $M(X, Y)$ for the pointwise multiplier space from X to Y , and $X \odot Y$ for the pointwise product space with its usual infimum product norm.

Define also the transported Hardy operator

$$H_\tau g(s) = \frac{1}{\tau(s)} \int_{[0, s]} |g(r)| d\tau(r),$$

where the integral is taken with respect to the Lebesgue–Stieltjes measure generated by the increasing function τ . Let

$$\mathcal{C}_\tau X = \{g : H_\tau g \in X\}, \quad \|g\|_{\mathcal{C}_\tau X} = \|H_\tau g\|_X.$$

When $\tau(s) = s$, this is the usual Cesaro optimal domain $\mathcal{C}X$.

Idea of the Proof

The entire mechanism is the identity

$$T_w(fg) = (T_w f)(T_w g).$$

Thus T_w is not only a lattice isometry $X^{[w]} \rightarrow X$, but also respects pointwise products. Multiplier norms and product factorizations are therefore transported exactly through the same change of variables.

Theorem 1 (Same-weight transport of multipliers and products). *Let X and Y be Banach ideal spaces on I . Then $T_w f = f \circ \tau$ is a multiplicative lattice isometry from $X^{[w]}$ onto X , and it induces isometric identifications*

$$M(X^{[w]}, Y^{[w]}) = M(X, Y)^{[w]}$$

and

$$X^{[w]} \odot M(X^{[w]}, Y^{[w]}) \cong X \odot M(X, Y).$$

Consequently,

$$Y^{[w]} = X^{[w]} \odot M(X^{[w]}, Y^{[w]})$$

with constant C if and only if

$$Y = X \odot M(X, Y)$$

with the same constant C . In particular, isometric factorization is preserved.

Proof. The definition gives $\|T_w f\|_X = \|f\|_{X^{[w]}}$, so T_w is an onto lattice isometry. Since composition with τ respects pointwise products,

$$T_w(hf) = (T_w h)(T_w f).$$

Therefore

$$\begin{aligned} \|h\|_{M(X^{[w]}, Y^{[w]})} &= \sup_{\|f\|_{X^{[w]}} \leq 1} \|hf\|_{Y^{[w]}} \\ &= \sup_{\|T_w f\|_X \leq 1} \|T_w(hf)\|_Y \\ &= \sup_{\|g\|_X \leq 1} \|(T_w h)g\|_Y \\ &= \|T_w h\|_{M(X, Y)}. \end{aligned}$$

This proves the multiplier identification $M(X^{[w]}, Y^{[w]}) = M(X, Y)^{[w]}$ isometrically.

For products, take $u \in X^{[w]} \odot M(X^{[w]}, Y^{[w]})$. If $|u| \leq \sum_k |a_k b_k|$, with $a_k \in X^{[w]}$ and $b_k \in M(X^{[w]}, Y^{[w]})$, then

$$|T_w u| \leq \sum_k |(T_w a_k)(T_w b_k)|.$$

The product norm of $T_w u$ in $X \odot M(X, Y)$ is at most the same sum of products of norms. Applying the same argument to T_w^{-1} gives the reverse inequality, hence an isometric product identification.

Finally, applying T_w to $Y^{[w]}$ gives Y , and applying T_w to the weighted product gives $X \odot M(X, Y)$. Thus the two factorization statements are equivalent, with identical constants. $\square$

Corollary 1 (Common-weight abstract analogue). *Every factorization theorem of the form*

$$Y = X \odot M(X, Y)$$

immediately yields the common-weight transported theorem

$$Y^{[w]} = X^{[w]} \odot M(X^{[w]}, Y^{[w]}).$$

For example, the source paper's unweighted factorizations through L_p , Cesaro spaces, Tandori spaces, or mixed Cesaro–Tandori pairs all have same-weight transported versions whenever the weighted spaces are interpreted through the same W -change of variables.

Theorem 2 (Transport of the Cesaro side). *The same map $T_w f = f \circ \tau$ is an isometric lattice isomorphism*

$$T_w : \mathcal{C}(X^{[w]}) \longrightarrow \mathcal{C}_\tau X.$$

Moreover,

$$M(X^{[w]}, \mathcal{C}(X^{[w]})) = M(X, \mathcal{C}_\tau X)^{[w]}$$

isometrically, and

$$\mathcal{C}(X^{[w]}) = X^{[w]} \odot M(X^{[w]}, \mathcal{C}(X^{[w]}))$$

if and only if

$$\mathcal{C}_\tau X = X \odot M(X, \mathcal{C}_\tau X).$$

Proof. For $f \in L_0(I)$ and $g = T_w f = f \circ \tau$, the change of variables $t = \tau(r)$ gives

$$(Hf)(\tau(s)) = \frac{1}{\tau(s)} \int_0^{\tau(s)} |f(t)| dt = \frac{1}{\tau(s)} \int_{[0,s]} |g(r)| d\tau(r) = H_\tau g(s).$$

Therefore

$$\|f\|_{\mathcal{C}(X^{[w]})} = \|Hf\|_{X^{[w]}} = \|(Hf) \circ \tau\|_X = \|H_\tau(T_w f)\|_X,$$

which proves $T_w \mathcal{C}(X^{[w]}) = \mathcal{C}_\tau X$ isometrically. The multiplier and product statements then follow exactly as in the previous theorem, using again that T_w respects pointwise products. $\square$

Full-Push Obstruction

The second theorem explains why the same-weight result above is not yet a full solution of Section 8.b for Cesaro spaces. The ordinary Hardy operator does not commute with the W -transport unless $W(t)$ is linear. Instead it becomes H_τ . Thus the weighted Cesaro factorization problem is equivalent to a factorization problem for the optimal domain $\mathcal{C}_\tau X$.

This is a real obstruction rather than just a notational inconvenience. In the special case $w \equiv 1$, we have $\tau(s) = s$ and $\mathcal{C}_\tau X = \mathcal{C}X$, so the transported weighted-Cesaro factorization reduces to the source paper's previous Section 8.a question

$$\mathcal{C}X = X \odot M(X, \mathcal{C}X).$$

Consequently, a blanket full answer to the abstract weighted-Cesaro problem would already settle that unweighted open problem.

Limitations

This theorem is deliberately scoped. It does not solve weighted factorizations with different weights on the two sides, where a single multiplicative transport no longer identifies both spaces at once. It also does not handle questions where a further symmetrization is part of the construction, such as abstract Lorentz spaces $\Lambda_{X,w}$, nor does it identify the separate open multiplier space $M(Ces_M, Ces_N)$ for Orlicz function spaces from Section 8.e of the source paper. For Cesaro spaces it gives an exact reduction to $\mathcal{C}_\tau X$, not a proof that $\mathcal{C}_\tau X$ always factorizes through X .

Verification and Novelty Check

The proof is purely formal after the change of variables $T_w f = f \circ W^{-1}$. The source paper itself uses the same transport observation to identify X_w with X for rearrangement invariant X , but it does not appear to state the corresponding multiplier and product-factorization transport principle in Section 8.b.

Run-index searches for arXiv:2401.05526, generalized weighted spaces, and weighted factorization found no prior packet for this result. Web searches for phrases such as $M(X_w, Y_w)$, generalized weighted space factorization, and weighted Cesaro factorization found related weighted Cesaro/Copson literature, including Barza–Marcoci–Marcoci’s weighted Hardy factorizations, but no exact same-weight transport statement under this formulation. The full-push check also shows that the Cesaro side is blocked by the weighted Hardy optimal-domain factorization $\mathcal{C}_\tau X = X \odot M(X, \mathcal{C}_\tau X)$.

Human Review Recommendation

Treat this as a likely valid partial result answering the common-weight transport version of Section 8.b, together with a reduction of the Cesaro weighted case to a weighted Hardy optimal-domain factorization problem. It should not be counted as a full solution to all weighted-space problems raised in the source paper.

References

1. T. Kiwerski and J. Tomaszewski, *Arithmetic, interpolation and factorization of amalgams*, arXiv:2401.05526, 2024.
2. S. V. Astashkin and L. Maligranda, *Structure of Cesaro function spaces*, Indag. Math. (N.S.) 20 (2009), 329–379.
3. S. Barza, A. Marcoci and L. Marcoci, *Factorizations of weighted Hardy inequalities*, Bull. Braz. Math. Soc. 49 (2018), 915–932.